

ARYA ZADEBRAHIMI

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PERSONAL WEBPAGE

EDUCATION

Aug 2024

Master of Science in Bioinformatics

Sharif University of Technology

- Cumulative GPA of 3.9 (18.2 / 20)
- Selected coursework: Synthetic Biology, Machine Learning for Bioinformatics, Cell and Molecular Biology, Structural Bioinformatics, Computational Genomics, Reinforcement Learning

May 2021

Bachelor of Science in Software Engineering

Azad University

- Cumulative GPA of 3 (15.58 / 20)
- Selected coursework: Probability and Statistics, Artificial Intelligence, Data Mining, Algorithm Design

RESEARCH INTEREST

- Microbial Communities
- Metabolic Engineering
- Graph Neural Networks
- Reinforcement Learning

RESEARCH EXPERIENCE

Synthetic auxotrophic community design with maximum growth rate

- Constrained optimization
- Based on previous work (SteadyCom)
- Ensures robustness due to modular community
- High complexity

Increase by-product yield using a synthetic community

- Constrained optimization
- Based on previous work (SteadyCom)
- Ensures maximum yield and robustness due to division of labor
- Computationally expensive

Organism flux prediction using Graph neural networks

- A large disconnected graph consists of metabolic networks as small connected graphs
- Achieved a Root Mean Square Error of 4 between predicted flux and Flux Balance Analysis results
- Currently reducing loss by adding meaningful features to each reaction

PROJECTS

- Analysis of microarray gene expression data to identify genes involved in AML (Acute Myeloid Leukemia)
- Docking and optimizing the conformation of the Fruquintinib inhibitor molecule
- A transformer-based model to predict financial market price
- Single-cell RNA sequencing clustering and analysis

TEACHING ASSISTANT

Generative Model Design

- Assignment Design
- Test Question Design

Biomedical Image Processing

- Assignment Design
- Test Question Design

FURTHER EXPERIENCE

Aug 2022

Wet Lab Techniques and Molecular Biology

Iran University of Medical Sciences, Tehran

- Short-term internship
- Cell DNA extraction
- DNA detection using PCR
- Preparing medium and cell culture
- Cell counting using microscope

Oct 2017 – Dec 2018

Android Developer

Tehran

- Developed 5 separate applications
- Agile project management methodology

SKILLS

TOEFL: 104 (Reading 30, Listening 28, Speaking 23, Writing 23)

Programming: Python (pytorch, pytorch geometric, transformers, gymnasium, numpy, pandas, opencv), Julia, Java, R, Android, TCL, C#, React